

Environment

Air Pollution

Raven

Air Pollution



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Forest Burning in Sumatra

- Fires are set to clear forests for planting
- In Asia, agriculture and silviculture result in more air pollution than industry
- Burning results in severe air pollution and poor visibility
- Smoke is full of small particles that can lodge in lungs and cause illness/disease



ZUMA Press, Inc./Alamy Stock Photo

Our Atmosphere

- Atmospheric Composition
 - Nitrogen 78.08%
 - Oxygen 20.95%
 - Argon 0.93%
 - Carbon dioxide 0.04%
- Ecosystem services
 - Blocks UV radiation
 - Moderates the climate
 - Redistributes water in the hydrologic cycle



SPL/Science Source

Air Pollution

- Chemicals added to the atmosphere by natural events or human activities in high enough concentrations to be harmful
- Two categories:
 - Primary Air Pollutant
 - Harmful substance emitted **directly into the atmosphere**
 - Secondary Air Pollutant
 - Harmful substance formed in the atmosphere when a **primary air pollutant reacts with substances normally found in the atmosphere or with other air pollutants**

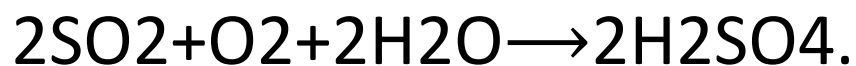
Major Air Pollutants (1 of 2)

Table 19.1 Major Air Pollutants

Pollutant	Composition	Primary or Secondary	Characteristics
Particulate Matter			
Dust	Variable	Primary	Solid particles
Lead	Pb	Primary	Solid particles
Sulfuric acid	H ₂ SO ₄	Secondary	Liquid droplets
Nitrogen Oxides			
Nitrogen dioxide	NO ₂	Primary	Reddish-brown gas
Sulfur Oxides			
Sulfur dioxide	SO ₂	Primary	Colorless gas with strong odor
Carbon Oxides			
Carbon monoxide	CO	Primary	Colorless, odorless gas
Carbon dioxide*	CO ₂	Primary	Colorless, odorless gas

How is sulfuric acid produced in the atmosphere?

Sulphuric acid is produced when sulphur dioxide reacts with oxygen and water in the presence of a catalyst according to the reaction:



sulfuric acid droplets can form in the air when sulfur dioxide is released from the burning of coal, oil, and gas.

Formation of nitrogen oxides in atmosphere

Nitrogen oxides are produced in **combustion processes**, partly from nitrogen compounds in the fuel, but mostly by direct combination of atmospheric oxygen and nitrogen in flames. Nitrogen oxides are produced naturally by lightning, and also, to a small extent, by **microbial processes in soils**.

Prompt NO_x is formed from molecular nitrogen in the air combining with **fuel in fuel-rich conditions** which exist, to some extent, in all combustion.

This nitrogen then oxidizes along with the fuel and becomes NO_x during combustion, just like fuel NO_x.

Formation of CO

Carbon monoxide is produced by **incomplete combustion, and also as a stage in the atmospheric oxidation of methane**. The chief natural sources of carbon monoxide are **wildfires**.

The other major source of carbon monoxide is combustion in **industrial, heating, and transport activities**.

Partial combustion in inefficient power plants and in inefficient heating systems (e.g., in east Asia) produces large amounts.

Formation of CO₂

Anthropogenic CO₂ sources include those from **power generation, transportation, industrial sources, chemical production, petroleum production, and agricultural practices**. Many of these source types burn fossil fuels (**coal, oil, and natural gas**), with CO₂ emissions as a byproduct.

Carbon dioxide is added to the atmosphere naturally when organisms **respire or decompose (decay), carbonate rocks are weathered, forest fires occur, and volcanoes erupt**. Carbon dioxide is also added to the atmosphere through human activities, such as the **burning of fossil fuels and forests** and the production of cement.

Major Air Pollutants (2 of 2)

Pollutant	Composition	Primary or Secondary	Characteristics
Hydrocarbons			
Methane	CH_4	Primary	Colorless, odorless gas
Benzene	C_6H_6	Primary	Liquid with sweet smell
Ozone	O_3	Secondary	Pale-blue gas with acrid odor
Hazardous Air Pollutants			
Chlorine	Cl_2	Primary	Yellow-green gas

a compound of hydrogen and carbon, such as any of those which are the chief components of **petroleum and natural gas**.

Hydrocarbons formation in atmosphere

Sometimes unburned hydrocarbons are released into the air during incomplete combustion. Burning fossil fuels, including gasoline in automobile engines, releases some hydrocarbons into the air. In a typical urban environment, the atmospheric concentration of hydrocarbons is around 3 ppm (parts per million).

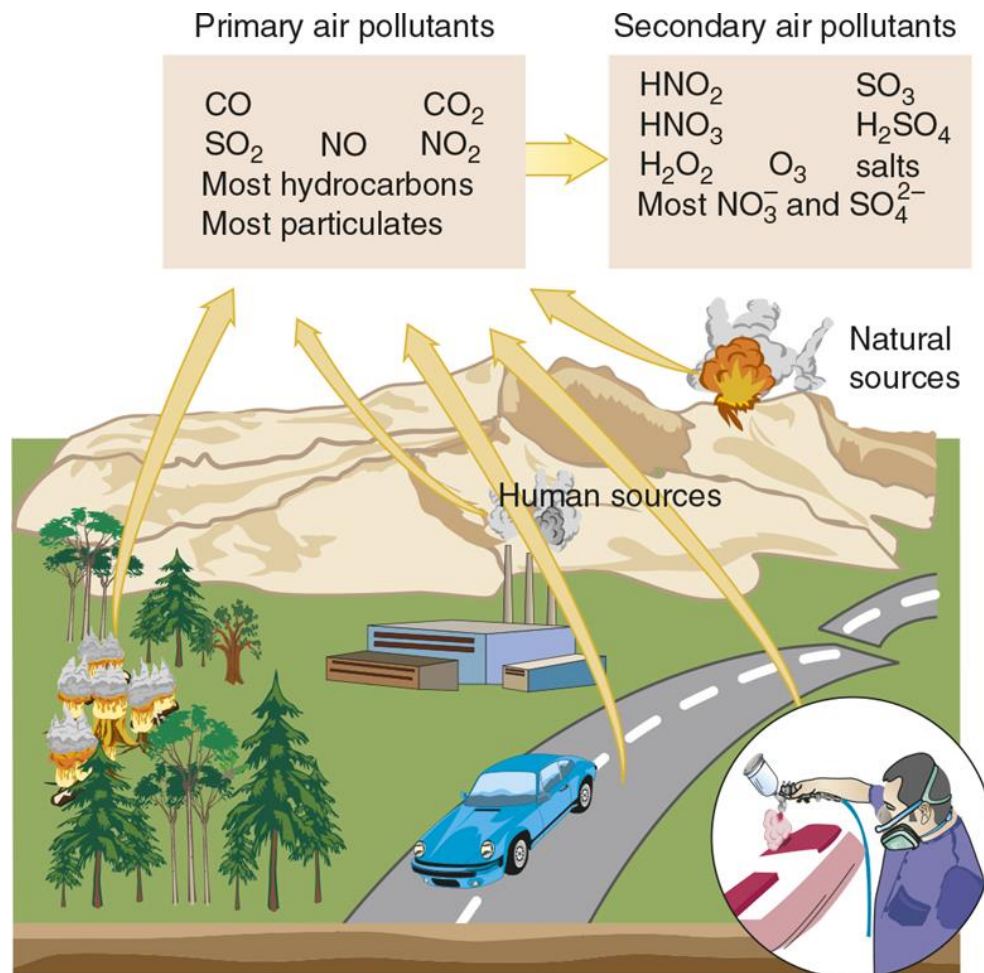
The production, storage, and transport of oil and natural gas are a major global source of hydrocarbons into the atmosphere. Coal, oil and natural gas all have hydrocarbons in them. They are mostly found deep underground, in porous rock formations, such as sandstone, limestone, and shale.

How does chlorine get into the air?

Chlorine gas can be released to the air when bleach is mixed with other **cleaning solutions that contain an acid**; for example, some toilet cleaners. Mixing bleach with ammonia also produces very hazardous gases, such as chloramines.

Possible sources of chlorine in the atmosphere are **industrial liquefaction processing, other industrial uses of chlorine, and accidental leakage** during storage or transportation. **Production of chlorine has doubled in the past 10 years** and is expected to continue at this rate of increase for several years.

Primary and Secondary Pollutants



Nitrous acid
 Nitric acid
 Sulfur trioxide
 Nitrate
 Sulfate
 Hydrogen peroxide
 Ozone

Major Classes of Air Pollutants

- Particulate Matter
- Lead
- Nitrogen Oxides
- Sulfur Oxides
- Carbon Oxides
- Hydrocarbons
- Ozone
- Hazardous Air Pollutants (HAPs)

Particulate Matter and Lead

- PM - thousands of different solid or liquid particles suspended in air
 - Includes: soil particles, soot, lead, asbestos, sea salt, and sulfuric acid droplets
 - May contain materials with toxic/carcinogenic effects
 - Small particles can become lodged in lungs
- Lead – soft metal used in industrial/chemical processes
 - Acute and chronic effects, including permanent impairment of cognitive ability

Soot: black flaky substance consisting of carbon, produced by the incomplete burning of organic matter.

Nitrogen and Sulfur Oxides

- Nitrogen Oxides
 - Gases produced by the chemical interactions between atmospheric nitrogen and oxygen at high temperature
 - Greenhouse gases that cause difficulty breathing
 - Sulfur Oxides
 - Gases produced by the chemical interactions between sulfur and oxygen
 - Causes acid precipitation
- Acid rain/deposition, includes any form of precipitation with acidic components, such as sulfuric or nitric acid that fall to the ground from the atmosphere in wet or dry forms. This can include rain, snow, fog, hail or even dust that is acidic.

Carbon Oxides and Hydrocarbons

- Carbon Oxides
 - Carbon monoxide (CO) – poisonous gas
 - Carbon dioxide (CO₂) – greenhouse gas
- Hydrocarbons
 - Diverse group of organic compounds that contain only hydrogen and carbon (ex: CH₄ - methane)
 - Some are related to **photochemical smog** and greenhouse gases

What is photochemical smog?

Photochemical smog is a mixture of pollutants that are formed when **nitrogen oxides and volatile organic compounds (VOCs)** react to sunlight, creating a brown haze above cities.

It tends to occur more often in summer, because that is when we have the most sunlight.

Ozone is the major contributor to photochemical smog.

Ozone

Ground-level ozone comes from pollution emitted from cars, power plants, industrial boilers, refineries, and chemical plants. Ozone pollution can even come from paints, cleaners, solvents, and motorized lawn equipment.

- Tropospheric Ozone
 - Man-made pollutant in the lower atmosphere
 - Secondary air pollutant
 - Component of photochemical smog
- Stratospheric Ozone
 - Essential component that screens out UV radiation in the upper atmosphere
 - Man-made pollutants (ex: CFCs) can destroy it

Ozone Damage to Grape Leaves

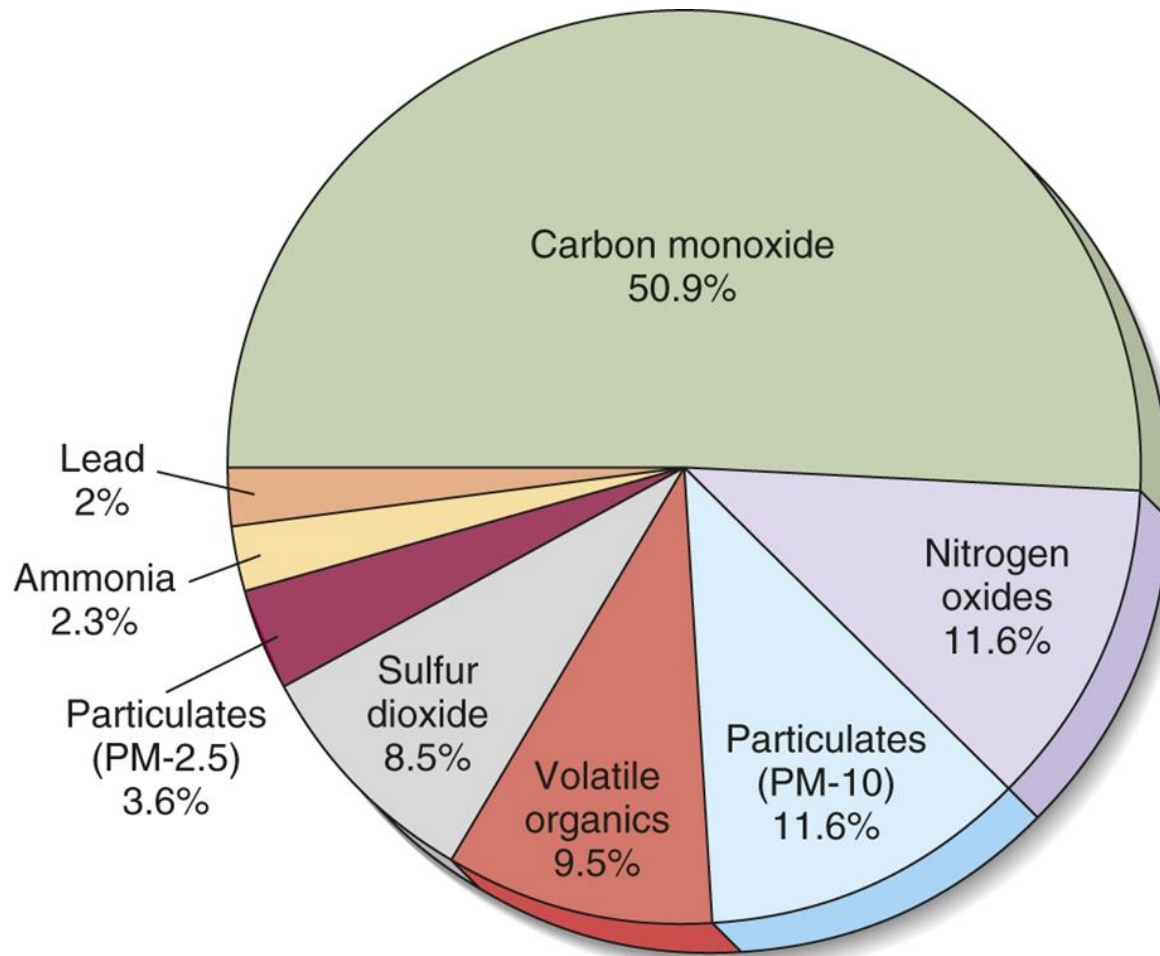


TED SPIEGEL/National Geographic
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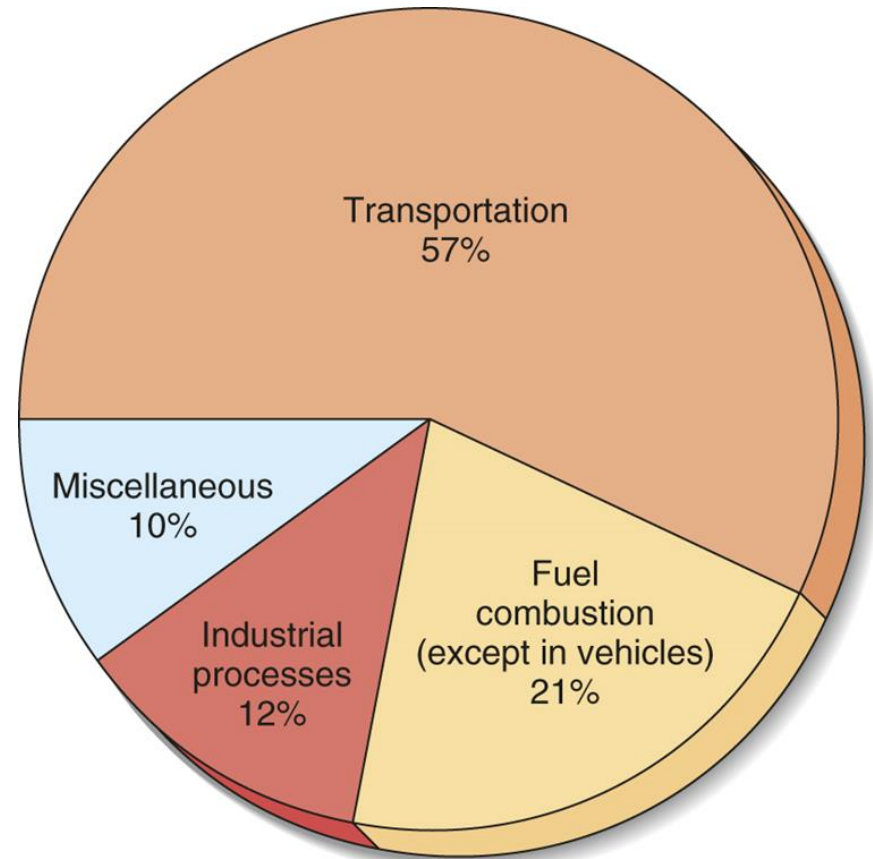
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Air Pollutants



Sources of Outdoor Air Pollution

- Two main sources:
 - Transportation
 - Industry
- Electric power plants and other stationary sources emit most particulate matter



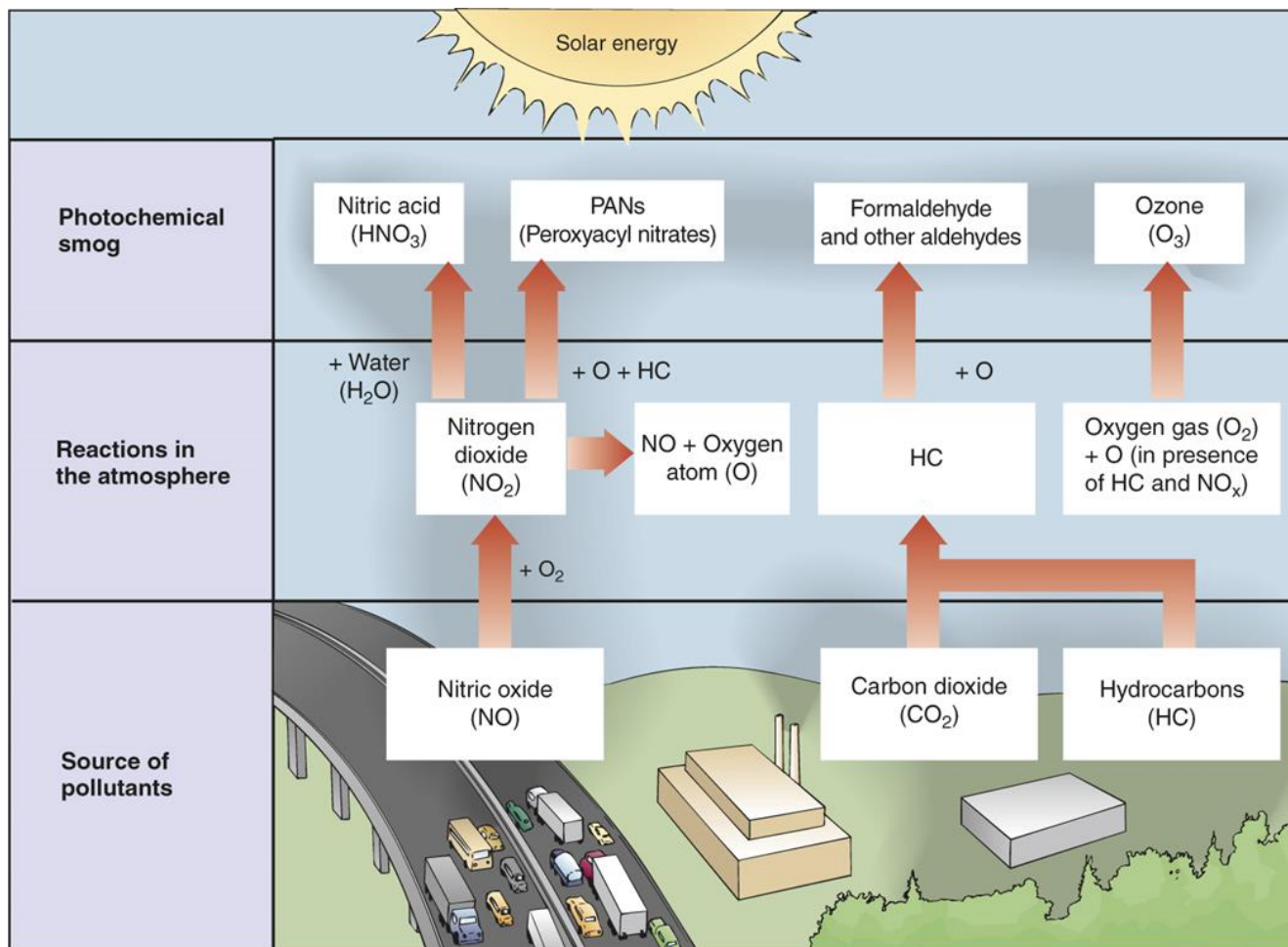
Urban Air Pollution

- Industrial smog – sulfur oxides and PM
- Photochemical smog (ex: Los Angeles below)
 - Brownish-orange haze formed by chemical reactions involving sunlight, nitrogen oxide, and hydrocarbons



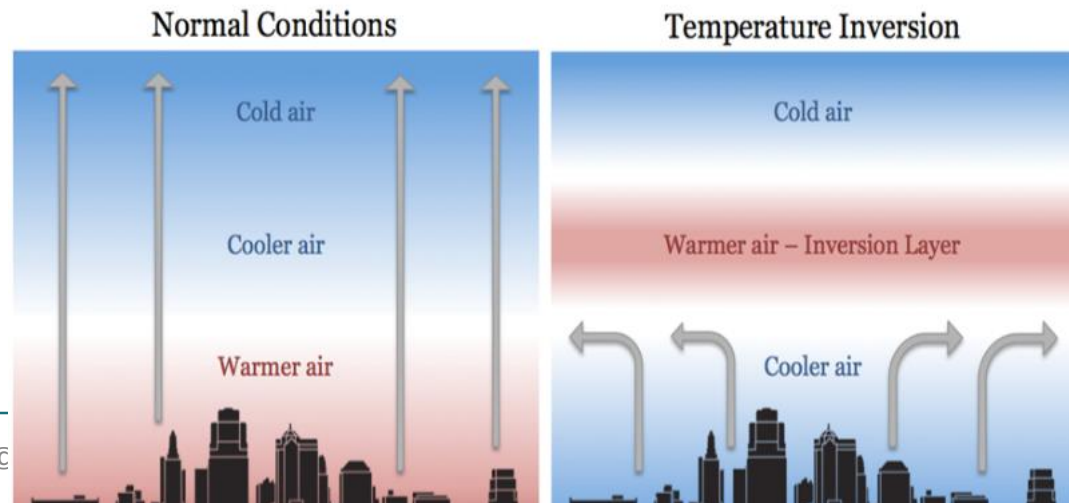
Jodi Cobb/National Geographic Creative

Formation of Photochemical Smog



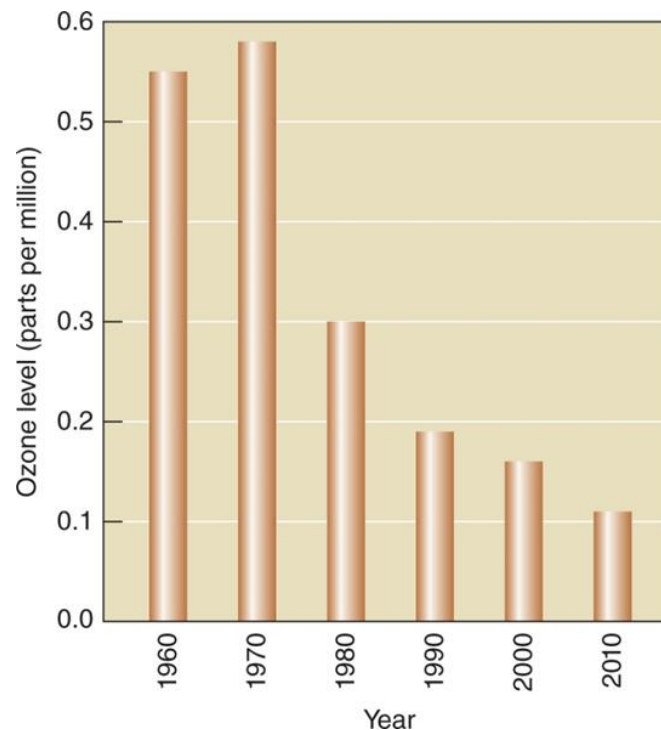
Efforts to Reduce Ozone in S. California

- Volatile organic compounds (VOCs)
 - Contributes to smog
 - Ex: compounds released from paints, cleaning products, dry cleaners, bakeries
- Temperature inversions
 - During inversion, temperature increases with altitude (usually reverse)
 - Pollution is trapped in high concentrations near surface and is not distributed

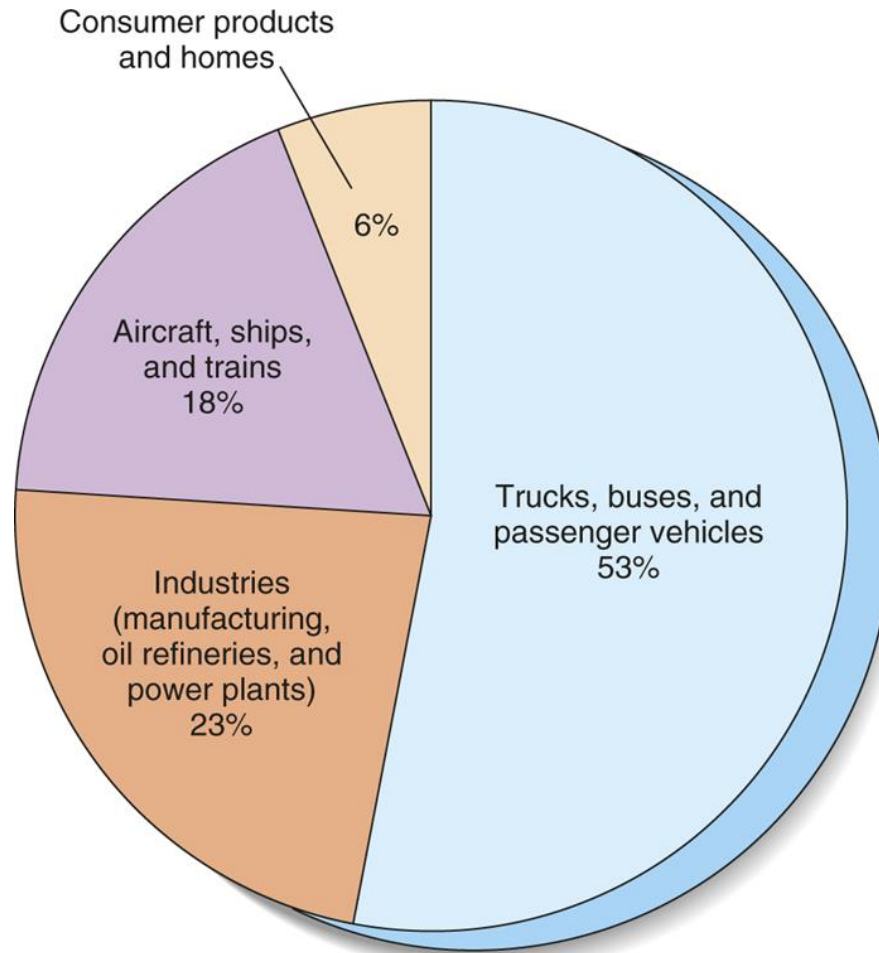


Peak Ozone in S. California

- CA consolidated efforts to improve air quality (1977)
- Improvements due to a few major regulations



Sources of Smog in Los Angeles



Effects of Air Pollution

- Low level exposure
 - Irritates eyes
 - Causes inflammation of respiratory tract
- Can develop into chronic respiratory diseases
 - Emphysema
 - Chronic bronchitis

In people with emphysema, the air sacs in the lungs (alveoli) are damaged. Over time, the inner walls of the air sacs weaken and rupture — creating larger air spaces instead of many small ones. This reduces the surface area of the lungs and, in turn, the amount of oxygen that reaches your bloodstream.

Bronchitis is a condition that develops when the airways in the lungs, called bronchial tubes, become inflamed and cause coughing, often with mucus production.

Health Effects of Pollutants

Table 19.2 Health Effects Of Several Major Air Pollutants

Pollutant	Source	Effects
Particulate matter	Industries, electric power plants, motor vehicles, construction, agriculture	Aggravates respiratory illnesses; long-term exposure may cause increased incidence of chronic conditions such as bronchitis; linked to heart disease; suppresses immune system; some particles, such as heavy metals and organic chemicals, may cause cancer or other tissue damage
Nitrogen oxides	Motor vehicles, industries, heavily fertilized farmland	Irritate respiratory tract; aggravate respiratory conditions such as asthma and chronic bronchitis
Sulfur oxides	Electric power plants and other industries	Irritate respiratory tract; same effects as particulates
Carbon monoxide	Motor vehicles, industries, fireplaces	Reduces blood's ability to transport oxygen; headache and fatigue at lower levels; mental impairment or death at high levels
Ozone	Formed in atmosphere (secondary air pollutant)	Irritates eyes; irritates respiratory tract; produces chest discomfort; aggravates respiratory conditions such as asthma and chronic Bronchitis

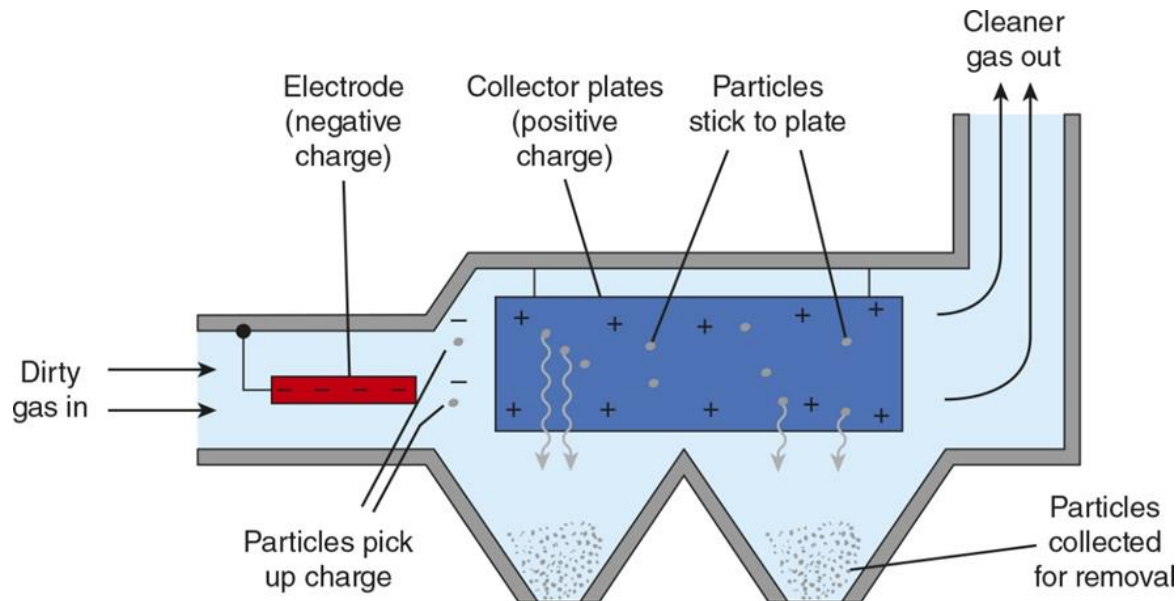
Children and Air Pollution

- Greater health threat to children than adults
 - Air pollution can restrict lung development
 - Children have higher breathing rate than adults
- Children who live in high ozone areas are more likely to develop asthma

Asthma is a chronic (long-term) condition that affects the airways in the lungs. The airways are tubes that carry air in and out of your lungs. If you have asthma, the airways can become inflamed and narrowed at times. This makes it harder for air to flow out of airways when we breathe out

Controlling Air Pollution

- Smokestacks with **electrostatic precipitator**
 - Electrode imparts negative charge on the air pollutants
 - Negatively charged pollutants are then attracted to positively charged walls- fall into collector



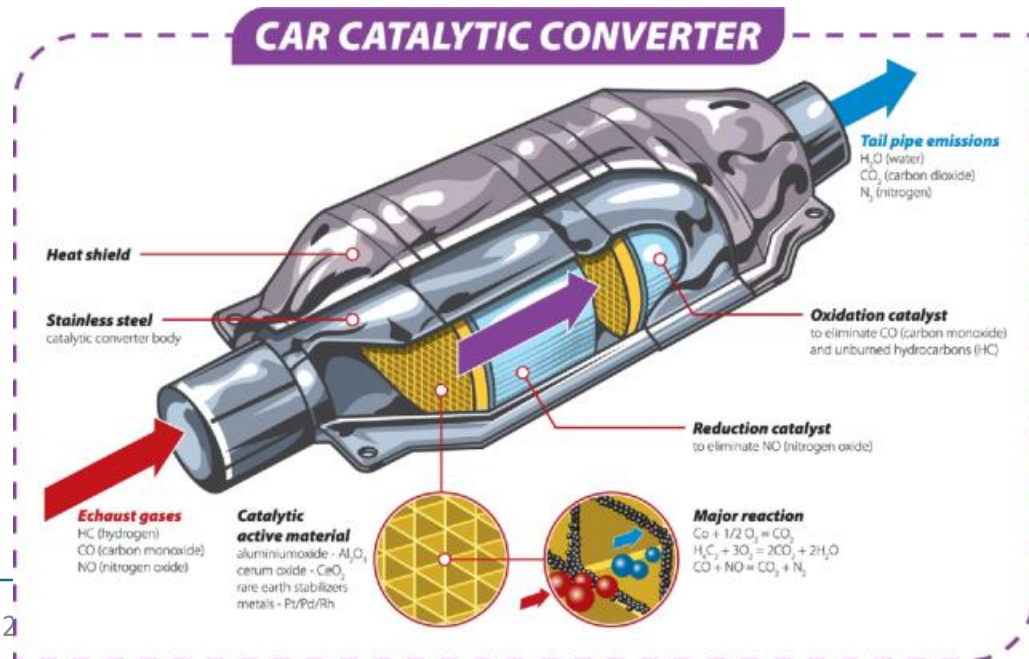
Land-excavation

- Particulate material can also be controlled by proper land-excavating techniques
 - Ex: sprinkling water on dry soil during road construction

Catalytic Converters How does it work?

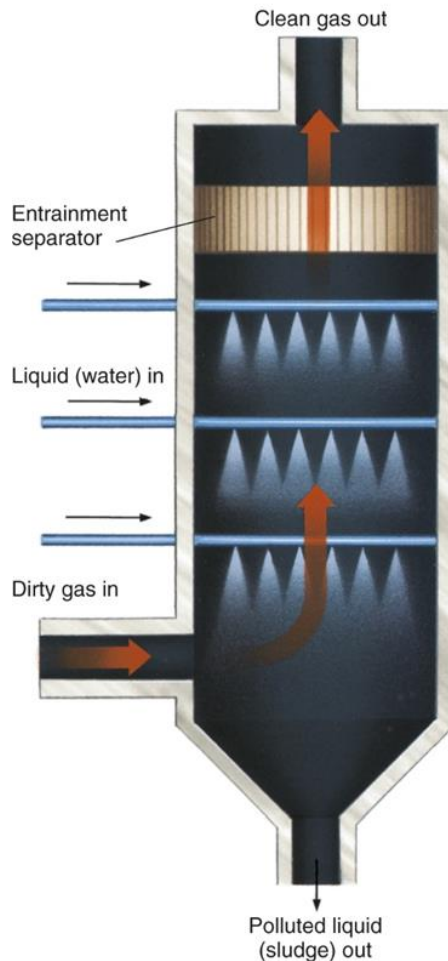
The catalytic converter works by using a **chemical reaction** to convert harmful pollutants into less harmful gases. It does this through a process called **catalysis**, which involves the use of a catalyst to speed up a chemical reaction. **When the exhaust gases pass through the converter, they come into contact with the catalyst, which causes the pollutants to break down into less harmful gases.**

First, the pollutants are absorbed onto the surface of the catalyst. Then, they react with oxygen in the air to form less harmful gases. E.g. carbon monoxide reacts with oxygen to form carbon dioxide, and nitrogen oxides react with oxygen to form nitrogen and oxygen gas. Finally, the less harmful gases are released into the atmosphere.



Smokestacks with Scrubbers

Entrainment- the process of making something part of a liquid or flow of something and carrying it along.



Scrubbers are an apparatus that cleans the gases passing through the smokestack of a coal-burning power plant.

Due to Clean Air Act regulations, most scrubbers in U.S. coal plants are used to remove sulfur emissions from coal and lessen the formation of acid rain.

Effect of Scrubbers



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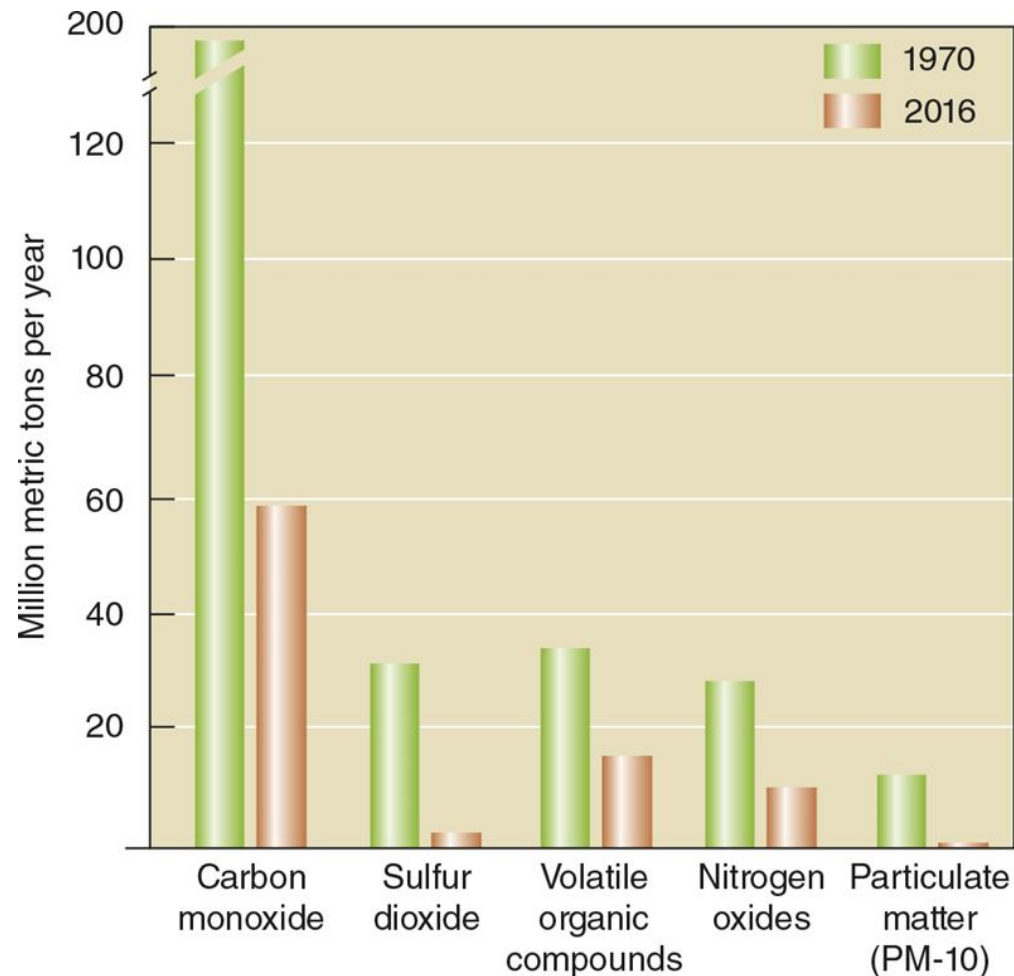
**Emissions not controlled-
heavily polluted (China)**



Harrison Shull/Aurora Photos

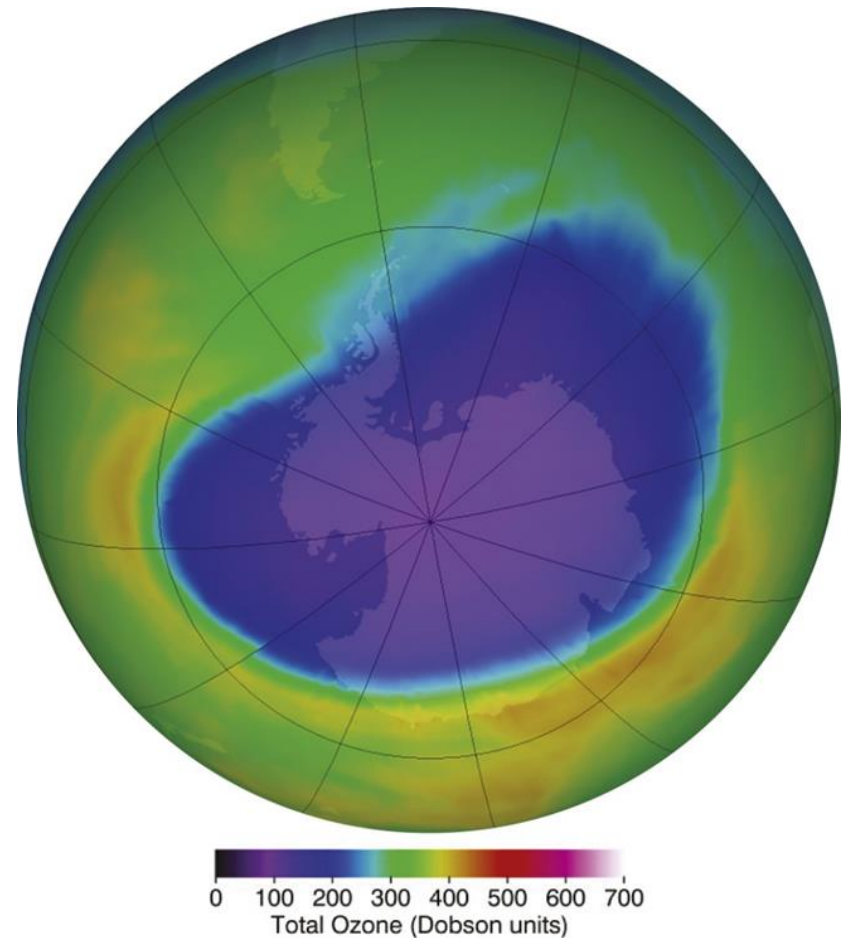
**Emissions controlled with
scrubbers-only steam is
expelled (North Carolina)**

Emissions in the United States



Ozone Thinning in Stratosphere

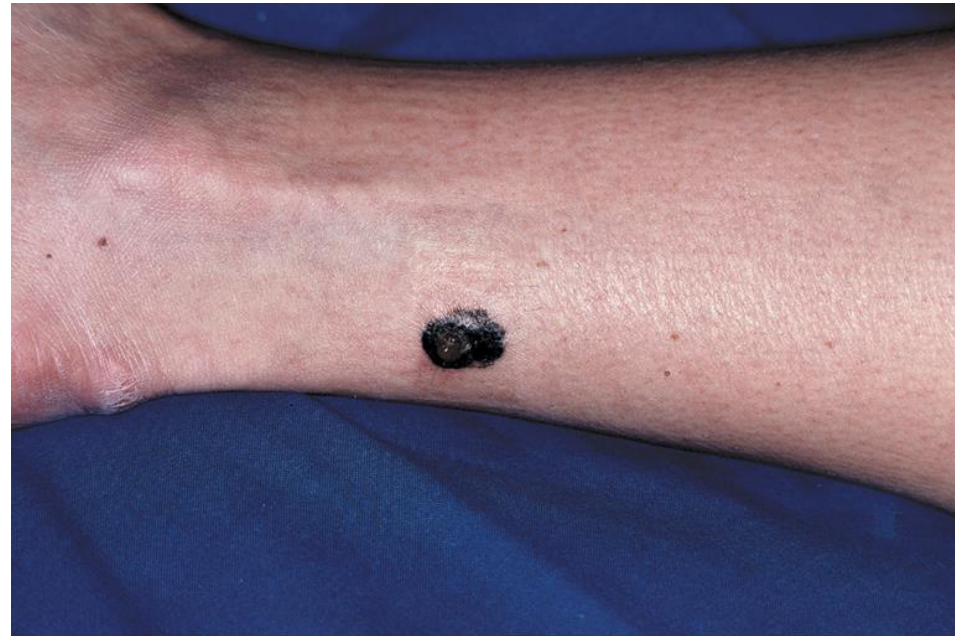
- Ozone thinning/hole
 - First identified in 1985 over Antarctica
 - Occurs annually between Sept. and Nov.
- Caused by human-produced bromine and chlorine containing chemicals (ex: CFCs)



NASA Goddard Space Flight Center

Effects of Ozone Depletion

- Higher levels of UV-radiation hitting the Earth
 - Eye cataracts
 - Skin cancer (right)
 - Weakened immunity
- May disrupt ecosystems
- May damage crops and forests



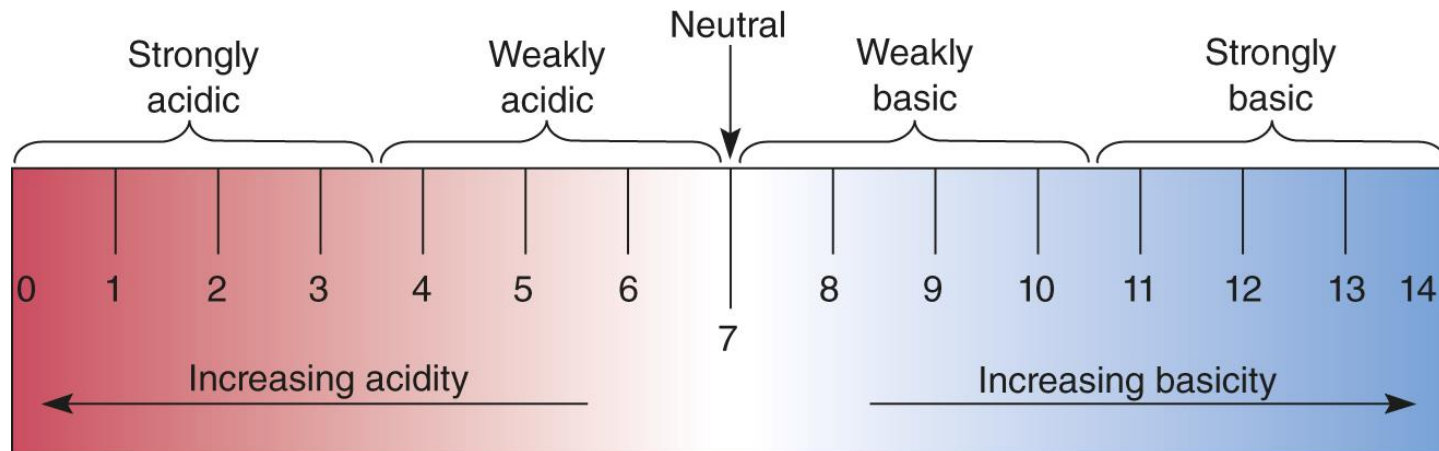
ISM/Medical Images

Recovery of Ozone Layer

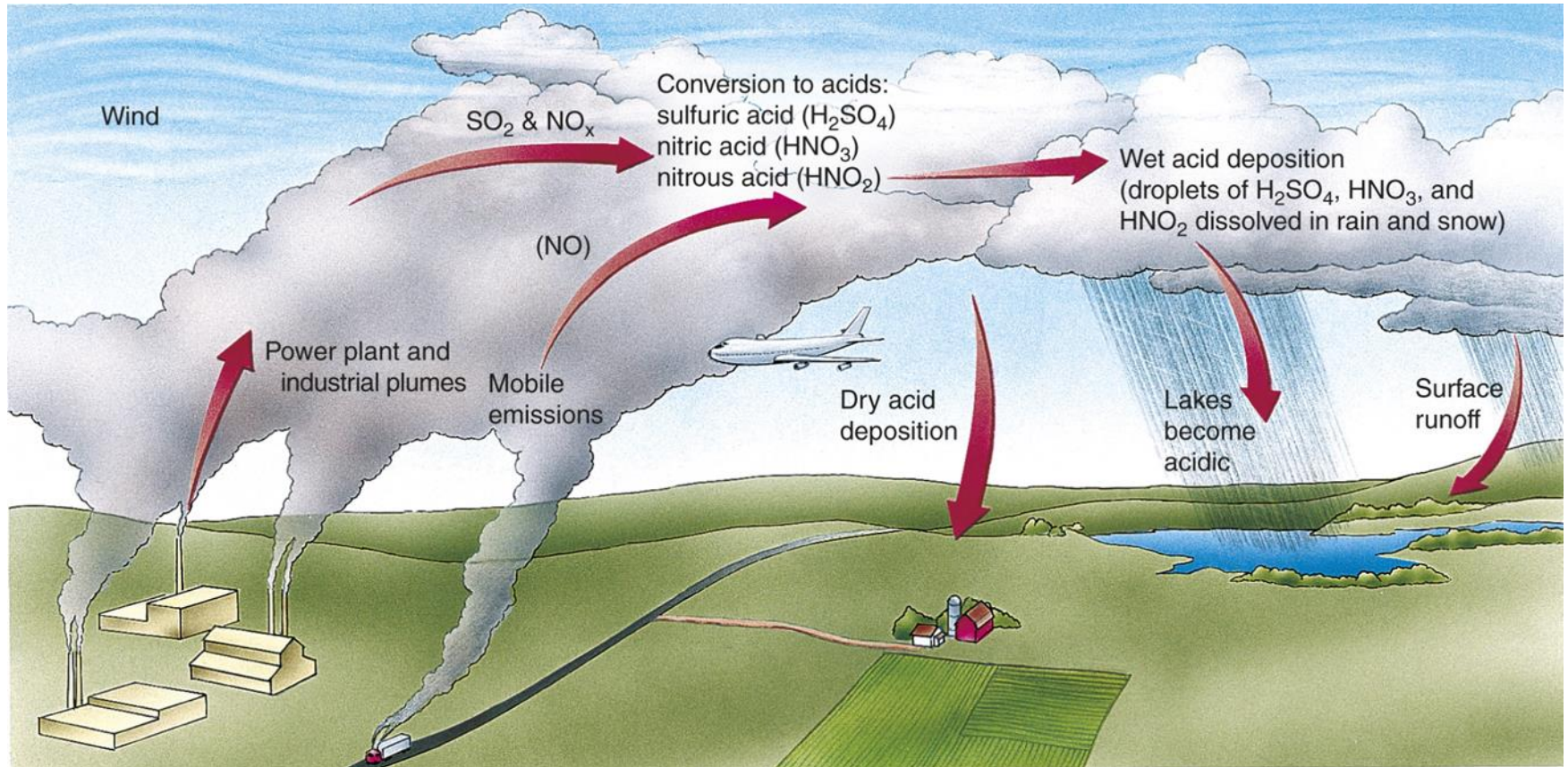
- Montreal Protocol (1987)
 - Reduction of CFCs
 - Started using HCFCs (greenhouse gas)
- All ozone destroying chemicals phased out in most countries
- Satellite pictures in 2000 indicated that ozone layer was recovering
- Full recovery will not occur until 2050

Acid Deposition

- Sulfur dioxide and nitrogen dioxide emissions react with water vapor in the atmosphere and form acids that return to the surface as either **dry** or **wet deposition**
- PH scale



How Acid Deposition Develops



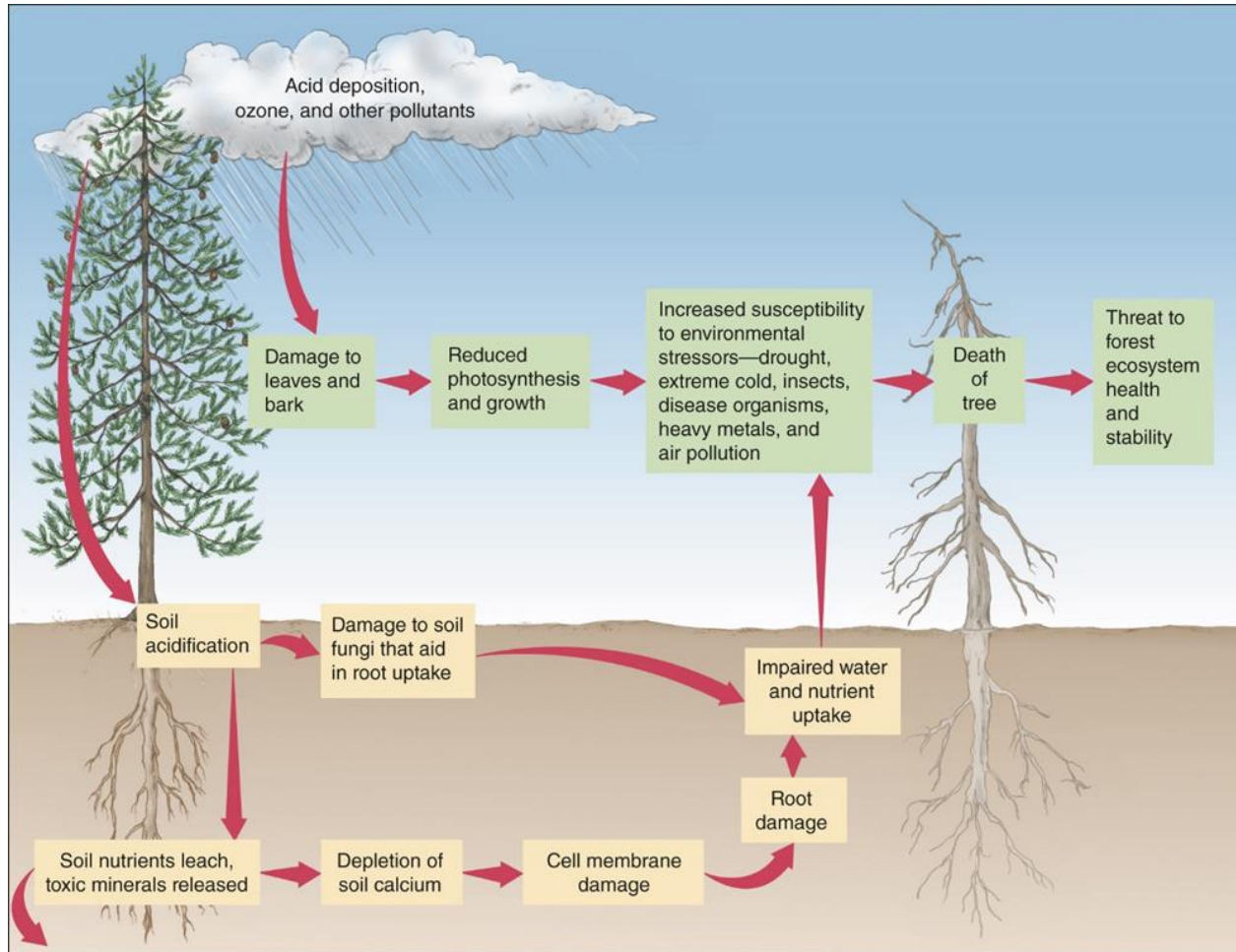
Effects of Acid Deposition

- Declining aquatic animal populations
- Thin-shelled eggs prevent bird reproduction
 - Calcium is unavailable in acidic soil
- Forest decline
 - Black Forest in Germany (right)



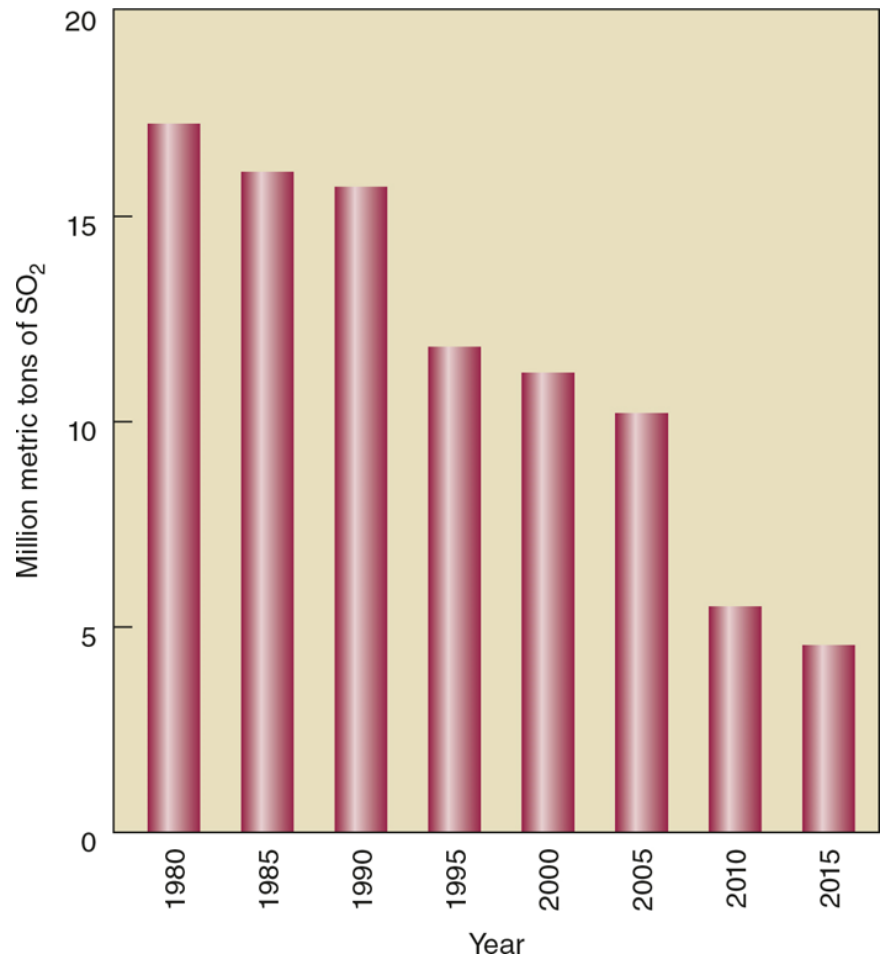
Hans Reinhard/Bruce Coleman, Inc./Photoshot

Forest Decline



Managing Acid Deposition

- Acid deposition is cross boundary issue
 - Pollution in one place, problem in another



Air Pollution Around the World

- Air quality is deteriorating rapidly in developing countries
 - Developing countries have older cars, air quality is low priority
- Liaoning Province, China
 - Residents only see sunlight a few weeks each year



SHENG LI/Reuters/Newscom

Air Pollution in Pakistan

A report said India and Pakistan experienced the worst air quality in the Central and South Asian regions, where nearly 60 percent of the population lives in areas where the concentration of PM2.5 particles is at least 7 times more than WHO's recommended levels.

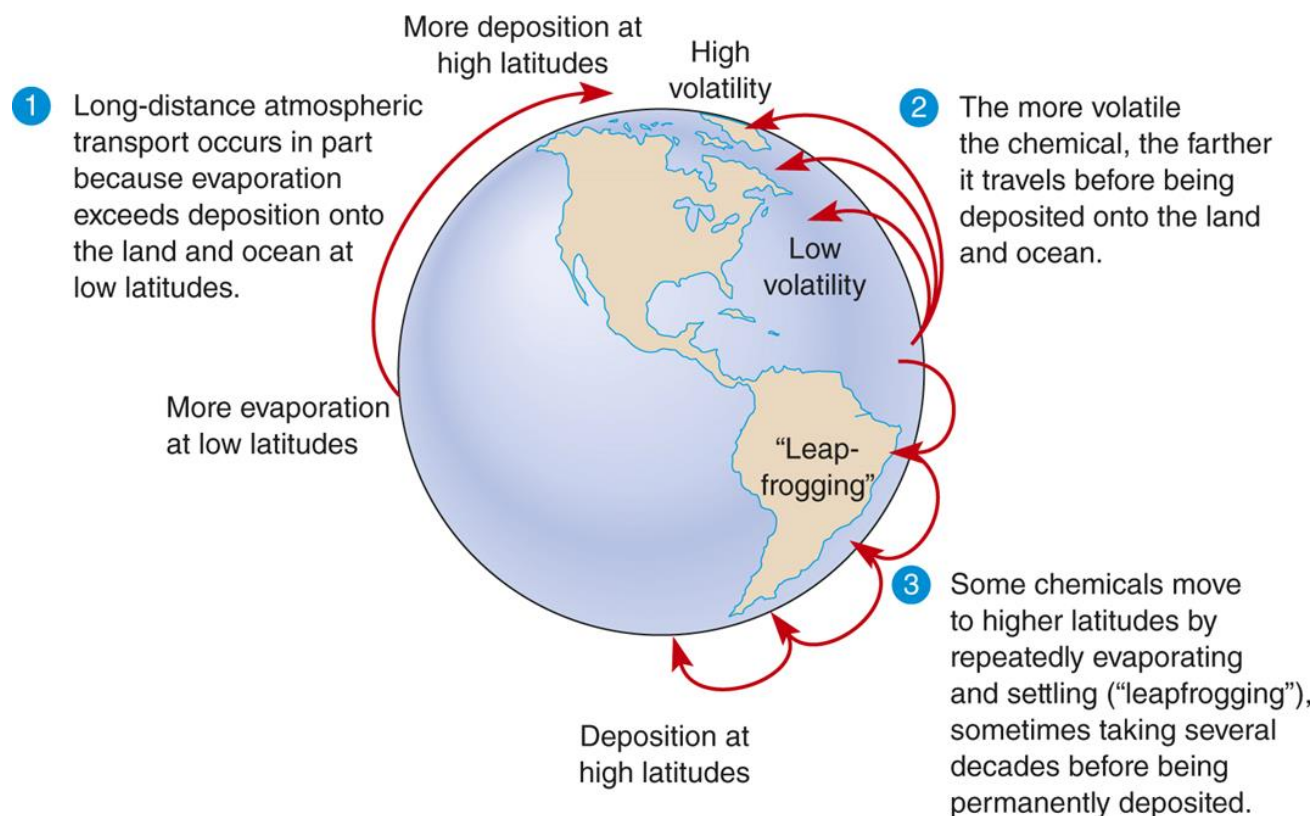


Adam Jones/Science Source



Long Distance Transport of Air Pollutants

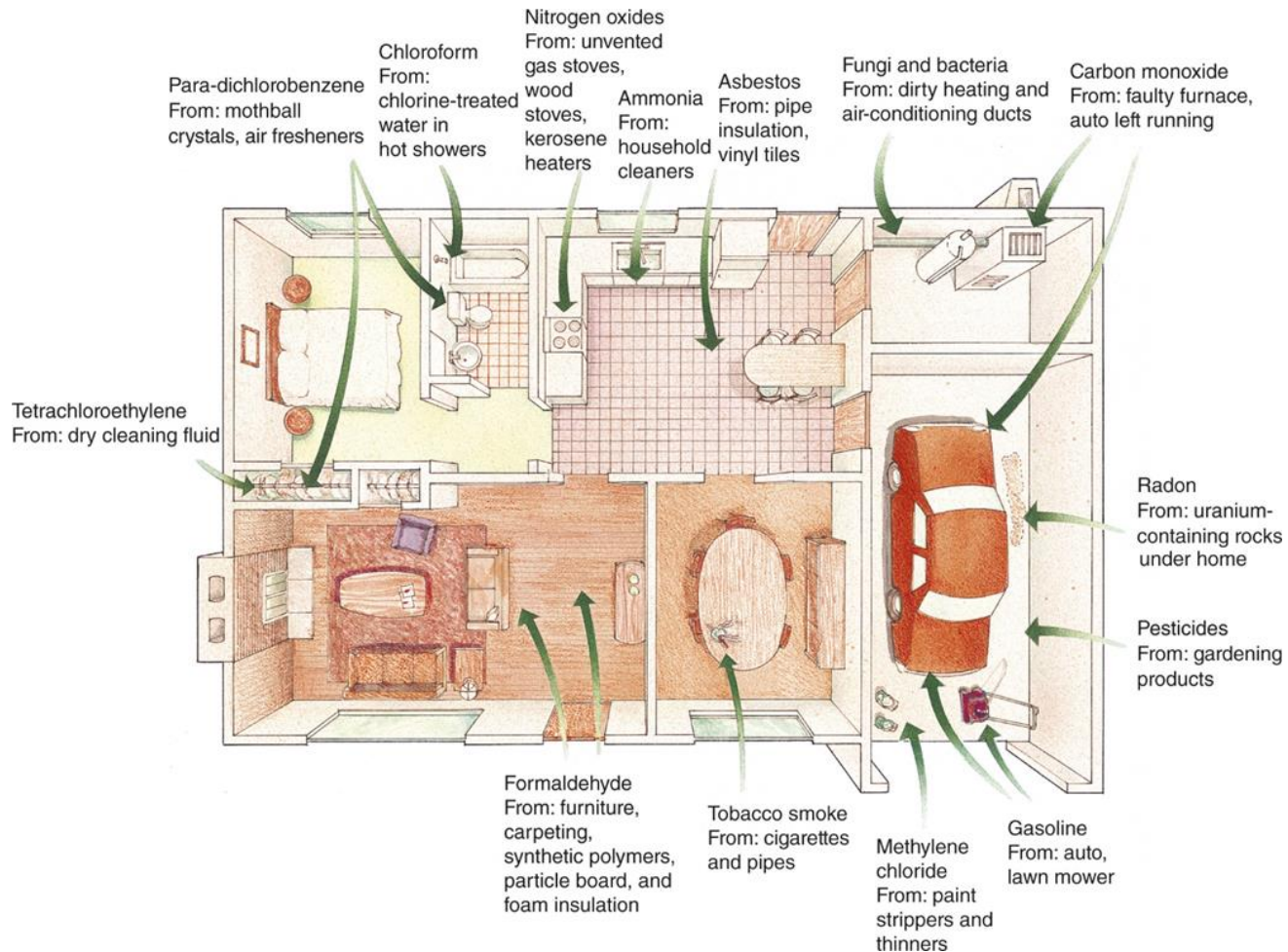
Global Distillation Effect



Sources of Household Air Pollution

- Pollutants can be 5–100x greater than outdoors
 - Many spend 95% of time indoors
- Most common contaminants:
 - Radon, cigarette smoke, carbon monoxide, nitrogen dioxide, formaldehyde, pesticides, lead, cleaning solvents, ozone, and asbestos

Indoor Air Pollution



Tobacco and Household Radon Infiltration

- Cigarette smoke
 - Increasing in developing nations
- Radon
 - Natural form of ionizing radiation
 - Damaging to surface tissues if not diluted with air
 - Trapped in houses

